

Enrico Magnago

APPLIED SCIENTIST · PHD · SOFTWARE DEVELOPER

enmag.github.io

EnMag

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enrico-magnago

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Inquisitive, self-motivated and perseverant problem solver with outstanding academic results, strong background in formal verification and industrial experience in large-scale distributed systems. Dependable and proficient software developer with experience in design, implementation and evaluation of symbolic model checking algorithms. Love to combine formal verification expertise and coding skills to build scalable verification tools and create reliable software products.

Expertise: Formal Verification, Temporal Logics, SAT, SMT, Timed / Hybrid Systems, Software Testing, Software Development, S3.

Experience

- SENIOR APPLIED SCIENTIST**
Amazon Web Services
Automated API learning, software verification and testing of large-scale distributed systems.
Berlin, Germany
2024/10/01 - ongoing
- APPLIED SCIENTIST II**
Amazon Web Services
Automated testing of large-scale distributed systems.
Berlin, Germany
2022/12/01 - 2024/09/30
- RESEARCH AND DEVELOPMENT ENGINEER**
Synopsys Inc
Design and development of tools for **static analysis** of interpreted programming languages.
Dublin, Ireland
2022/05/09 - 2022/10/31
- PHD STUDENT / CANDIDATE**
University of Trento, Fondazione Bruno Kessler
Significantly enhanced the **falsification** capabilities of model checkers in **infinite-state** and **timed systems**.
Designed an effective **SMT-based symbolic algorithm** capable of finding counterexamples that other state-of-the-art procedures miss.
Developed a novel **compositional** semi-automated approach to falsify temporal properties. item Evaluated the viability of cloud platforms (**AWS, Azure**) for performance testing.
Published **5 research papers**: 3 in highly rated international conferences and 2 in technical journals.
Trento, Italy
2018/11/01 - 2022/11/18
- TEACHING ASSISTANT: FORMAL METHODS COURSE, MASTER IN COMPUTER SCIENCE**
University of Trento
Gave the laboratory lectures, defined the exam exercises and evaluated the students' solutions.
Topics: model checking using **Spin, NuSMV** and **nuXmv**; verification of timed systems using **timed-nuXmv** and **HyCOMP**.
Trento, Italy
2019/03/15 - 2019/12/31
- SOFTWARE DEVELOPER**
Fondazione Bruno Kessler, Embedded Systems (ES) unit
Extended the symbolic model checker nuXmv to support verification of **timed systems**.
Designed and implemented techniques to verify expressive specifications: **Metric Temporal Logic**.
Defined and evaluated a novel and effective approach to identify infinite **non-lasso counterexamples** for timed systems.
Developed a **new compiler module** in nuXmv to meet software re-usability and extensibility requirements.
Testing, bug fixing, refactoring and documentation of **large code base**:
>1M lines of **C**, **>200K** lines of **Python**, **>150K** lines of **C++**, **>50k** lines of **LaTeX**.
Trento, Italy
2017/03/27 - 2018/10/31
- INTERN**
Fondazione Bruno Kessler, Data & Knowledge Management (DKM) unit
Implemented a Semantic Web application database interface using **Elasticsearch** Java API.
Trento, Italy
2015/07/20 - 2015/10/20

Education

- PHD, INFORMATION AND COMMUNICATION TECHNOLOGY**
Fondazione Bruno Kessler, University of Trento
Cum Laude.
Advisor: Alessandro Cimatti, Co-advisor: Alberto Griggio.
Trento, Italy
2018/11/01 - 2022/11/18
- SUMMER / WINTER SCHOOLS ABOUT FORMAL VERIFICATION**
Technical University of Munich, University of Verona, University of Lisbon
MOD 2019: Summer School on Safety and Security of Software Systems: Logics, Proofs, Applications.
CPS 2019: Summer School on Formal Methods for Cyber-Physical Systems.
VMCAI 2019: Winter School on Verification, Model Checking, and Abstract Interpretation.
Marktoberdorf, Verona, Lisbon
2019/07/31 - 2019/08/09
2019/06/03 - 2019/06/07
2019/01/09 - 2019/01/12
- MASTER DEGREE IN COMPUTER SCIENCE**
University of Trento
Final grade: 110L/110, Average grade: 29.24/30.
Notable project: distributed consensus protocol in Go.
Trento, Italy
2016/08/08 - 2018/10/10

- **POST-GRADUATE ERASMUS STUDENT**
University of Edinburgh, School of Informatics
· Notable project: algorithms using Hadoop Map-Reduce.

Edinburgh, United Kingdom
2016/09/05 - 2016/12/24

- **BACHELOR DEGREE IN COMPUTER SCIENCE**
University of Trento
· Final grade: 110L/110, Average grade: 28.11/30.
· Notable project: multi-process question-answer game in C.

Trento, Italy
2013/08/08 - 2016/07/18

- **HIGH SCHOOL DIPLOMA**
Liceo Bertrand Russell
· Final grade: 93/100.

Cles, Italy
2008/09/19 - 2013/07/04

Programming Languages

C	●●●●●	NuSMV ¹ , nuXmv ¹ .
C++	●●●●●	nuXmv ¹ , msatic3 ¹ .
Python	●●●●●	F3 ² , pySMT ¹ , General Scripting.
Java	●●●●●	Development of large scale services.
Rust	●●●●●	Professional use.
Scala	●●●●●	Professional use.
LaTeX	●●●●●	Publications, Tools Documentation, BSc, MSc and PhD Theses.
Lex-Yacc	●●●●●	NuSMV ¹ , nuXmv ¹ , Academic Project.
sh	●●●●●	General Scripting.

1: Pre-existent tool or library
2: Standalone software

Skills

Languages: ●●●●● Italian ●●●●● English
Hobbies: ●●●●● Trumpet ●●●●● Skiing

Publications

OSDI 2026	High Fidelity Models for Large Scale Stateful Services Nouraldin Jaber, Dongyun Jin, Bernhard Kragl, Enrico Magnago, Gustavo Petri, Thorsten Tarrach, and Serdar Tasir <i>20th USENIX Symposium on Operating Systems Design and Implementation.</i>
INFOCOMP 2022	LTL falsification in infinite-state systems Alessandro Cimatti, Alberto Griggio, and Enrico Magnago. <i>Information and Computation Journal, Volume 289.</i>
ATVA 2021	Automatic Discovery of Fair Paths in Infinite-State Transition Systems Alessandro Cimatti, Alberto Griggio, and Enrico Magnago. <i>19th International Symposium on Automated Technology for Verification and Analysis.</i>
VMCAI 2021	Proving the Existence of Fair Paths in Infinite-State Systems Alessandro Cimatti, Alberto Griggio, and Enrico Magnago. <i>22th International Conference on Verification, Model Checking, and Abstract Interpretation.</i>
INFOCOMP 2020	SMT-based satisfiability of first-order LTL with event freezing functions and metric operators Alessandro Cimatti, Alberto Griggio, Enrico Magnago, Marco Roveri, and Stefano Tonetta. <i>Information and Computation Journal, Volume 272.</i>
CAV 2019	Extending nuXmv with Timed Transition Systems and Timed Temporal Properties Alessandro Cimatti, Alberto Griggio, Enrico Magnago, Marco Roveri, and Stefano Tonetta. <i>31st International Conference on Computer Aided Verification.</i>

Authors are in alphabetical order.

Professional Activities

2025	CAV Program Committee. <i>International Conference on Computer Aided Verification.</i>
2025 - 2026	FMCAD Program Committee. <i>International Conference on Formal Methods in Computer-Aided Design.</i>
2025 - 2026	OOPSLA Artifact Evaluation Committee. <i>Conference on Object-Oriented Programming Systems, Languages, and Applications.</i>
2020, 2022 - 2025	CAV Artifact Evaluation Committee. <i>International Conference on Computer Aided Verification.</i>
2024	PLDI Artifact Evaluation Committee. <i>Symposium on Programming Language Design and Implementation.</i>
2024 - 2025	ATVA Artifact Evaluation Committee. <i>International Symposium on Automated Technology for Verification and Analysis.</i>
2022	FORMATS Artifact Evaluation Committee. <i>International Conference on Formal Modeling and Analysis of Timed Systems.</i>